Project Report: Complexity Science and Emergent Dynamics

Topic: Pressure Avalanches and Self-Organized Criticality in Microchannel Networks

Submitted by:

Sumit

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Under the Guidance of:

Professor Rajesh Khanna

Department of Chemical Engineering

Indian Institute of Technology Delhi

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Abstract

This project explores the application of Complexity Science to the phenomenon of pressure-driven avalanches in microchannel networks. By utilizing the principles of Self-Organized Criticality (SOC), we simulate a porous medium (microchannel grid) subjected to continuous fluid flow and fouling. The study focuses on how localized clogging events (noise) lead to non-linear pressure redistributions (avalanches) that follow a power-law distribution. The results demonstrate that microchannel networks naturally evolve toward a critical state where systemic failures (large-scale clogging) are emergent properties of the network's connectivity rather than isolated incidents. This study provides insights into the resilience and management of industrial micro-systems.

1 Introduction

Complexity science provides a revolutionary framework for understanding systems where the "whole is greater than the sum of its parts." In traditional engineering, we often look at systems in isolation or through linear approximations. However, real-world systems like social networks, power grids, and microfluidic channels exhibit emergent behaviors that arise from the local interactions of individual components.

This project specifically targets the behavior of fluid transport in micro-scale conduits, a field that is critical for the pharmaceutical and fine-chemical industries. The unpredictability of fouling (clogging) in these channels represents a classic complex systems problem.

2 Application of Complexity Science (Requirement 4a)

Microchannel networks are ideal candidates for complexity science for several reasons:

- **Non-linearity:** A minor deposit in one pore can drastically alter the pressure head across the entire network.
- **Lack of Central Control:** There is no "global controller" deciding where a particle will settle. The flow distribution is determined entirely by local pressure gradients.
- **High Interconnectivity:** Channels are often fed by a common manifold. If one channel clogs, the fluid is forced into neighboring channels, increasing their load.

- **Phase Transitions:** The system can suddenly shift from a "smooth flow" state to a "total blockage" state through a series of cascades.

3 System Description (Requirement 4b)

To model this system, we use a 2D cellular automaton (simulated in NetLogo) with the following characteristics:

3.1 Noise (Perturbation)

In our model, "noise" represents the continuous operation of a pump. At every time step, fluid containing microscopic foulants is introduced. Mathematically, we increment the 'pressure' of a randomly selected patch ($P_{i,j}$) by 1. This represents the accumulation of stress or particles in a single pore.

3.2 Connectivity

The system connectivity is defined by the physical arrangement of pores. We use a `neighbors4` (von Neumann) connectivity, where each pore shares its pressure with its immediate North, South, East, and West neighbors. This represents the interconnected nature of porous media or micro-manifolds.

3.3 The Avalanche (Redistribution)

The "avalanche" is the core emergent event. When the pressure in a single pore exceeds a critical threshold ($T > 3$), it "bursts" or flushes.

$$P_{i,j} \to P_{i,j} - 4 \tag{1}$$

The excess pressure is then distributed to the four neighbors:

$$P_{neighbors} \to P_{neighbors} + 1 \tag{2}$$

This can cause neighbors to exceed their thresholds, triggering a massive cascade—a pressure avalanche.

4 Computer Simulation (Requirement 4c)

The simulation was developed in **NetLogo 6.4**. The environment consists of a grid of patches where color coding represents the pressure intensity.

4.1 Algorithm Logic

The simulation follows a loop where noise is added, and then a `while` loop processes the avalanches until the system returns to a sub-critical state. This ensures that the time-scale of the avalanche is much faster than the time-scale of the noise, which is a requirement for observing SOC.

5 Analysis of Frequency Distribution (Requirement 4d)

A critical component of this project is analyzing the size and frequency of the avalanches.

5.1 Frequency vs. Size

We recorded the size of every avalanche (number of pores that burst in one cascade) over 10,000 ticks.

The data shows a clear **Power Law** distribution:

$$N(s) = C \cdot s^{-\tau} \tag{3}$$

where s is the size and $N(s)$ is the frequency. On a log-log scale, this appears as a straight line. This confirms that the system does not have a "typical" avalanche size; instead, it exhibits scale-invariance.

5.2 Relation to Connectivity

If we increase the connectivity (e.g., using `neighbors8`), the system becomes more resilient to small avalanches but produces much larger "global" avalanches. In chemical engineering terms, a highly connected manifold prevents local clogging but risks a total system blowout if the critical threshold is reached everywhere.

6 Self-Organized Critical State (Requirement 4e)

The simulation demonstrates that the microchannel network naturally "organizes" itself into a critical state. We did not have to tune the pressure to a specific value. The constant "noise" of the pump pushes the system toward the edge of instability, while the avalanches pull it back.

The system lives at the "Edge of Chaos." This explains why industrial microchannels often work perfectly for hours and then suddenly fail without warning. The failure

is not caused by a "big particle" but by the internal state of the system being at a critical point where even a tiny addition of pressure can trigger a system-wide collapse.

7 Conclusion

The application of complexity science to microfluidic networks reveals that fouling and pressure distribution are not random independent events. They are part of a self-organized process. For engineering design, this implies that "perfectly uniform" systems might be the most dangerous, as they can lead to synchronized, catastrophic avalanches. A certain level of heterogeneity or "firebreaks" in the network might be necessary for resilience.

A NetLogo Source Code

```
;; --- Complete NetLogo Code for Pressure Avalanches ---
globals [ avalanche-size avalanche-history ]
patches-own [ pressure ]

to setup
  clear-all
  set avalanche-history []
  ask patches [
    set pressure random 4
    update-color
  ]
  reset-ticks
end

to go
  set avalanche-size 0
  ask one-of patches [ set pressure pressure + 1 ]
  redistribute-flow
  if avalanche-size > 0 [
    set avalanche-history lput avalanche-size avalanche-history
  ]
  tick
end

to redistribute-flow
  let overloaded patches with [pressure > 3]
  while [any? overloaded] [
    ask overloaded [
      set pressure pressure - 4
      set avalanche-size avalanche-size + 1
      ask neighbors4 [ set pressure pressure + 1 ]
    ]
    ask patches [ update-color ]
    set overloaded patches with [pressure > 3]
  ]
end

to update-color
  ifelse pressure = 0 [ set pcolor black ] [
```

```
    ifelse pressure = 1 [ set pcolor 85 ] [
      ifelse pressure = 2 [ set pcolor 65 ] [
        ifelse pressure = 3 [ set pcolor 15 ] [
          set pcolor white
        ]
      ]
    ]
  ]
end
```

B Prompts and AI Assistance Acknowledgement

I acknowledge the use of generative AI tools for assisting in the generation of NetLogo code structures and the formatting of this LaTeX report. The conceptual framework was derived from the "Complexity Science in Chemical Industry" course.